

Continuous Mapping

Theorem 1.1. Suppose (X_n) is a sequence of random variables and g is a continuous mapping, then:

- i. If $X_n \xrightarrow{as} X$, then $g(X_n) \xrightarrow{as} g(X)$.
- ii. If $X_n \xrightarrow{p} X$, then $g(X_n) \xrightarrow{p} g(X)$.
- iii. If $X_n \xrightarrow{d} X$, then $g(X_n) \xrightarrow{d} g(X)$.

□

Proof. (i.) Suppose $X_n \xrightarrow{as} X$, then there exists a set $A \subseteq \Omega$ such that:

$$\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)$$

for $\forall \omega \in A$ and $\mathbb{P}(A) = 1$. Since g is a continuous function, on the same set:

$$g\{X(\omega)\} = g\left\{\lim_{n \rightarrow \infty} X_n(\omega)\right\} = \lim_{n \rightarrow \infty} g\{X_n(\omega)\}.$$

Conclude that $g(X_n) \xrightarrow{as} g(X)$.

(ii.) Suppose $X_n \xrightarrow{p} X$, and let $m \in \mathbb{N}$. By total probability:

$$\begin{aligned} \mathbb{P}(|g(X_n) - g(X)| > \epsilon) &= \mathbb{P}(|g(X_n) - g(X)| > \epsilon, |X_n - X| < m^{-1}) \\ &\quad + \mathbb{P}(|g(X_n) - g(X)| > \epsilon, |X_n - X| > m^{-1}). \end{aligned}$$

Proposition 1.1.

$$\lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \mathbb{P}(|g(X_n) - g(X)| > \epsilon, |X_n - X| < m^{-1}) = 0.$$

◆

Define:

$$A_{mn} = \{\omega : |g(X_n) - g(X)| > \epsilon, |X_n - X| < m^{-1}\}, \quad A_m = \bigcup_{n=1}^{\infty} A_{mn}.$$

Observe first that the sequence (A_m) is decreasing, since if $m_2 > m_1$ and $\omega \in A_{m_2}$ then $\omega \in A_{m_1}$. Fix ω , and hence $X(\omega)$. By continuity of g at $X(\omega)$, there exists δ such that if $|\xi - X(\omega)| < \delta$ then $|g(\xi) - g(X(\omega))| < \epsilon$. Choose M such that $M^{-1} < \delta$, then $\omega \notin A_m$ for $m \geq M$. Since ω was arbitrary:

$$\bigcap_{m=1}^{\infty} A_m = \emptyset.$$

Consequently,

$$\lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \mathbb{P}(A_{mn}) \leq \lim_{m \rightarrow \infty} \mathbb{P}(A_m) = \mathbb{P}\left(\bigcap_{m=1}^{\infty} A_m\right) = 0.$$

Proposition 1.2.

$$\lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \mathbb{P}(|g(X_n) - g(X)| > \epsilon, |X_n - X| > m^{-1}) = 0.$$

◆

This follows from $X_n \xrightarrow{p} X$ since:

$$\begin{aligned} \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \mathbb{P}(|g(X_n) - g(X)| > \epsilon, |X_n - X| > m^{-1}) \\ \leq \lim_{m \rightarrow \infty} \left\{ \lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > m^{-1}) \right\} = \lim_{m \rightarrow \infty} \{0\} = 0. \end{aligned}$$

Overall, if $X_n \xrightarrow{p} X$, then for $\forall \epsilon > 0$:

$$\lim_{n \rightarrow \infty} \mathbb{P}(|g(X_n) - g(X)| > \epsilon) = 0.$$

(iii.) Suppose $X_n \xrightarrow{d} X$. Let F_n denote the distribution of X_n and F the distribution of X . Also, let G_n denote the distribution of $g(X_n)$ and G the distribution of $g(X)$. The aim is to show that $G_n \rightarrow G$ (at points of continuity of G). By the Skorokhod representation theorem, there exist (ξ_n) and ξ , defined on a common probability space, such that ξ_n has distribution F_n , ξ has distribution F , and $\xi_n \xrightarrow{as} \xi$. Now, by (i.) $g(\xi_n) \xrightarrow{as} g(\xi)$, and by the convergence hierarchy, $g(\xi_n) \xrightarrow{as} g(\xi)$ implies $g(\xi_n) \xrightarrow{d} g(\xi)$. Since $g(\xi_n)$ has distribution G_n and $g(\xi)$ has distribution G , conclude $G_n \rightarrow G$ (at points of continuity of G). ■

Slutsky's

Remark 2.1. Recall from the portmanteau theorem that the following statements are equivalent:

- i. $X_n \xrightarrow{d} X$,
- ii. $\mathbb{E}\{g(X_n)\} \rightarrow \mathbb{E}\{g(X)\}$ for every bounded, continuous function g .
- iii. $\mathbb{E}\{\mathcal{L}(X_n)\} \rightarrow \mathbb{E}\{\mathcal{L}(X)\}$ for every bounded, Lipschitz function $\mathcal{L}$.

◆

Proposition 2.1. If $X_n \xrightarrow{d} X$ and $\|Y_n - X_n\| = o_p(1)$, then $Y_n \xrightarrow{d} X$.

◆

Proof. Let $\mathcal{L}$ denote a bounded, Lipschitz function. Then, there $\exists$ a constant L such that $\|\mathcal{L}(x) - \mathcal{L}(y)\| \leq L\|x - y\|$. Moreover, since $\mathcal{L}$ is bounded, $\|\mathcal{L}(x)\| \leq M$. First consider:

$$\begin{aligned} \|\mathbb{E}\{\mathcal{L}(Y_n)\} - \mathbb{E}\{\mathcal{L}(X_n)\}\| &\leq \mathbb{E}\|\mathcal{L}(Y_n) - \mathcal{L}(X_n)\| \\ &= \mathbb{E}\{\|\mathcal{L}(Y_n) - \mathcal{L}(X_n)\| \mathbb{I}(\|Y_n - X_n\| < \epsilon)\} \\ &\quad + \mathbb{E}\{\|\mathcal{L}(Y_n) - \mathcal{L}(X_n)\| \mathbb{I}(\|Y_n - X_n\| \geq \epsilon)\} \\ &\leq L\mathbb{E}\{\|Y_n - X_n\| \mathbb{I}(\|Y_n - X_n\| < \epsilon)\} + 2M\mathbb{P}\{\|Y_n - X_n\| \geq \epsilon\} \\ &< L\epsilon\mathbb{P}\{\|Y_n - X_n\| < \epsilon\} + 2M\mathbb{P}\{\|Y_n - X_n\| \geq \epsilon\}. \end{aligned}$$

Next, consider:

$$\begin{aligned} \|\mathbb{E}\{\mathcal{L}(Y_n)\} - \mathbb{E}\{\mathcal{L}(X)\}\| &\leq \|\mathbb{E}\{\mathcal{L}(Y_n)\} - \mathbb{E}\{\mathcal{L}(X_n)\}\| + \|\mathbb{E}\{\mathcal{L}(X_n)\} - \mathbb{E}\{\mathcal{L}(X)\}\| \\ &< L\epsilon + 2M\mathbb{P}\{\|Y_n - X_n\| \geq \epsilon\} + \|\mathbb{E}\{\mathcal{L}(X_n)\} - \mathbb{E}\{\mathcal{L}(X)\}\|. \end{aligned}$$

Taking the limit as $n \rightarrow \infty$, the second and third terms vanish since $\|Y_n - X_n\| = o_p(1)$ and $X_n \xrightarrow{d} X$, hence:

$$\lim_{n \rightarrow \infty} \|\mathbb{E}\{\mathcal{L}(Y_n)\} - \mathbb{E}\{\mathcal{L}(X)\}\| < K\epsilon.$$

Since $\epsilon > 0$ was arbitrary, the limit vanishes. Conclude that $Y_n \xrightarrow{d} X$. ■

Proposition 2.2. Suppose $X_n \xrightarrow{d} X$ and Y_n converges in probability to a constant α ($Y_n \xrightarrow{p} \alpha$), then (X_n, Y_n) converge jointly in distribution to (X, α) . ◆

Proof. Let $g(x, y)$ denote a bounded, continuous function. Define $h(x) = f(x, \alpha)$. Since $X_n \xrightarrow{d} X$, by the portmanteau theorem, $\mathbb{E}\{h(X_n)\} \rightarrow \mathbb{E}\{h(X)\}$, and equivalently:

$$\mathbb{E}\{g(X_n, \alpha)\} \rightarrow \mathbb{E}\{g(X, \alpha)\}.$$

Thus $(X_n, \alpha) \xrightarrow{d} (X, \alpha)$. Now $\|(X_n, Y_n) - (X_n, \alpha)\| = \|Y_n - \alpha\| = o_p(1)$. Together, $(X_n, \alpha) \xrightarrow{d} (X, \alpha)$ and $\|(X_n, Y_n) - (X_n, \alpha)\| = o_p(1)$ imply $(X_n, Y_n) \xrightarrow{d} (X, \alpha)$. ■

Theorem 2.1 (Slutsky's). If $X_n \xrightarrow{d} X$ and Y_n converges in probability to a constant α ($Y_n \xrightarrow{p} \alpha$), then for a continuous mapping f :

$$f(X_n, Y_n) \xrightarrow{d} f(X, \alpha).$$

□

Proof. By the preceding propositions, $(X_n, Y_n) \xrightarrow{d} (X, \alpha)$, and by the continuous mapping theorem $f(X_n, Y_n) \xrightarrow{d} f(X, \alpha)$. ■

Law of Large Numbers

Theorem 3.1 (LLN). Suppose (X_i) is a sequence of independent random variables. If the following conditions hold:

- i. $\lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbb{P}(|X_i| > n) = 0$,
- ii. and $\lim_{n \rightarrow \infty} n^{-2} \sum_{i=1}^n \mathbb{E}\{X_i^2 \mathbb{I}(|X_i| \leq n)\} = 0$,

then,

$$\text{plim}_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n (X_i - \mu_{n,i}) = 0,$$

where $\mu_{n,i} = \mathbb{E}\{X_i \mathbb{I}(|X_i| \leq n)\}$. □

Proof. Define $Y_{n,i} = X_i \mathbb{I}(|X_i| \leq n)$, and:

$$S_n = \sum_{i=1}^n X_i, \quad T_n = \sum_{i=1}^n Y_{n,i}.$$

Proposition 3.1. The original sum S_n and the truncated sum T_n converge to the same limit: $S_n - T_n = o_p(1)$. ◆

This follows from hypothesis (i.) since:

$$\begin{aligned} \mathbb{P}(|S_n - T_n| > \epsilon) &\leq \mathbb{P}(S_n \neq T_n) \leq \mathbb{P}\left(\bigcup_{i=1}^n \{X_i \neq Y_{n,i}\}\right) \\ &\leq \sum_{i=1}^n \mathbb{P}(X_i \neq Y_i) = \sum_{i=1}^n \mathbb{P}(|X_i| > n) \rightarrow 0. \end{aligned}$$

Proposition 3.2.

$$\text{plim}_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n (Y_{n,i} - \mu_{n,i}) = 0.$$
◆

This follows from hypothesis (ii.) since:

$$\begin{aligned} \mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n (Y_{n,i} - \mu_{n,i})\right| \geq \epsilon\right) &\leq \frac{1}{\epsilon^2 n^2} \text{Var}(T_n) \leq \frac{1}{\epsilon^2 n^2} \sum_{i=1}^n \mathbb{E}(Y_{n,i}^2) \\ &\leq \frac{1}{\epsilon^2 n^2} \sum_{i=1}^n \mathbb{E}\{X_i^2 \mathbb{I}(|X_i| \leq n)\} \rightarrow 0. \end{aligned}$$

Overall:

$$\text{plim}_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n (X_i - \mu_{n,i}) = \text{plim}_{n \rightarrow \infty} \frac{S_n - T_n}{n} + \text{plim}_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n (Y_{n,i} - \mu_{n,i}) = 0.$$
■

Example 3.1. Let (X_i) denote a sequence of independent random variables, with mean $\mathbb{E}(X_i) = \mu_i$ and $\text{Var}(X_i) = \sigma_i^2$. Suppose:

$$\mu = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu_i$$

exists, and that the variances are uniformly bounded $\text{Var}(X_i) \leq M$. Then:

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mu.$$

To see this, let $Y_i = X_i - \mu_i$ such that $\mathbb{E}(Y_i) = 0$ for $\forall i \in \mathbb{N}$, then:

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbb{P}(|Y_i| > n) \leq \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{\text{Var}(Y_i)}{n^2} \leq \lim_{n \rightarrow \infty} \frac{n \cdot M}{n^2} = 0.$$

The second condition follows similarly:

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{i=1}^n \mathbb{E}\{Y_i^2 \mathbb{I}(|Y_i| \leq n)\} \leq \lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{i=1}^n \text{Var}(Y_i) = \lim_{n \rightarrow \infty} \frac{n \cdot M}{n^2} = 0.$$

♠

Theorem 3.2 (Kolmogorov LLN). Suppose (X_n) are IID. There exists a constant α such that:

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{as} \alpha$$

if and only if $\mathbb{E}(X_i) < \infty$, in which case $\alpha = \mathbb{E}(X_i)$. □

Remark 3.1. See Resnick (2014), theorem 7.5.1; Serfling (1980), section 1.8. ♦

3.1 Glivenko-Cantelli

Theorem 3.3. Suppose (X_i) are IID random variables with common distribution function F . Let $\mathbb{F}_n$ denote the empirical distribution function:

$$\mathbb{F}_n(x, \omega) = \frac{1}{n} \sum_{i=1}^n I\{X_i(\omega) \leq x\}.$$

Then $\mathbb{F}_n$ converges to F uniformly almost surely:

$$\sup_{x \in \mathbb{R}} |\mathbb{F}_n(x, \omega) - F(x)| \xrightarrow{as} 0.$$

□

Remark 3.2. See Resnick (2014), theorem 7.5.2. ♦

Central Limit Theorem

4.1 IID Case

Proposition 4.1 (Lindeberg-Levy CLT). If (Y_i) are IID random variables with $\mathbb{E}(Y_i) = 0$ and $\mathbb{E}(Y_i^2) = 1$,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i \xrightarrow{d} N(0, 1).$$

◆

Proof. Consider the characteristic function of $n^{-1/2} \sum_{i=1}^n Y_i$:

$$\phi_n(t) = \mathbb{E}(e^{it \cdot n^{-1/2} \sum_{i=1}^n Y_i}) = \{\phi(n^{-1/2}t)\}^n = \left\{ \phi(0) + \frac{t}{n^{1/2}} \dot{\phi}(0) + \frac{t^2}{2n} \ddot{\phi}(0) + o(n^{-1}) \right\}^n,$$

where $\phi(\cdot)$ is the characteristic function of Y . Now:

$$\dot{\phi}(0) = i\mathbb{E}(Y) = 0, \quad \ddot{\phi}(0) = -\mathbb{E}(Y^2) = -1.$$

Thus:

$$\phi_n(t) = \left\{ \phi(0) - \frac{t^2}{2n} + o(n^{-1}) \right\}^n \rightarrow e^{-t^2/2}.$$

Since $\exp(-t^2/2)$ is the characteristic function of the standard normal distribution, by Levy's continuity theorem, $n^{-1/2} \sum_{i=1}^n Y_i$ converges in distribution to $N(0, 1)$. ■

Theorem 4.1 (Berry Essen). Suppose (X_i) are IID with mean $\mathbb{E}(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2 > 0$ and $\mathbb{E}|X_i|^3 < \infty$. Let F_n denote the finite-sample distribution of the normalized sum:

$$Z_n = \sqrt{n} \frac{\sum_{i=1}^n (X_i - \mu)}{\sigma},$$

then there exists a constant $C > 0$ such that for $\forall n \in \mathbb{N}$:

$$\sup_{x \in \mathbb{R}} |F_n(x) - \Phi(x)| \leq \frac{C \mathbb{E}|X_i - \mu|^3}{\sigma^3 \sqrt{n}},$$

where Φ is the standard normal distribution function. □

Remark 4.1. See Serfling (1980), section 1.9. ◆

4.2 Lindeberg Feller

Theorem 4.2 (Lindeberg-Feller CLT). Let (X_i) denote a sequence of independent random variables. Define the centered random variables $Y_i = X_i - \mathbb{E}(X_i)$. Suppose $\text{Var}(X_i) = \mathbb{E}(Y_i^2) = \sigma_i^2 < \infty$. Let $s_n^2 = \sum_{i=1}^n \sigma_i^2$. If, for $\forall \epsilon > 0$, the **Lindeberg condition** holds:

$$\lim_{n \rightarrow \infty} \frac{1}{s_n^2} \sum_{i=1}^n \mathbb{E}\{Y_i^2 \mathbb{I}(|Y_i| > \epsilon s_n)\} = 0, \quad (A_1)$$

then:

$$\frac{\sum_{i=1}^n (X_i - \mu_i)}{\sqrt{(\sum_{i=1}^n \sigma_i^2)}} \xrightarrow{d} N(0, 1). \quad (B)$$

□

Remark 4.2. See Resnick (2014), theorem 9.8.1. For the multivariate extension, see Serfling (1980), section 1.9. ♦

Proposition 4.2. The Lindeberg condition (A_1) implies:

$$\lim_{n \rightarrow \infty} \frac{\max_{1 \leq i \leq n} \sigma_i^2}{\sum_{i=1}^n \sigma_i^2} = 0. \quad (A_2)$$

♦

Proof.

$$\begin{aligned} \frac{\sigma_i^2}{s_n^2} &= \frac{\mathbb{E}(Y_i^2)}{s_n^2} = \frac{1}{s_n^2} \mathbb{E}\{Y_i^2 \mathbb{I}(|Y_i| \leq \epsilon s_n)\} + \frac{1}{s_n^2} \mathbb{E}\{Y_i^2 \mathbb{I}(|Y_i| > \epsilon s_n)\} \\ &\leq \epsilon^2 + \frac{1}{s_n^2} \sum_{i=1}^n \mathbb{E}\{Y_i^2 \mathbb{I}(|Y_i| \leq \epsilon s_n)\}. \end{aligned}$$

Since this bound is independent of i :

$$\frac{\max_{1 \leq i \leq n} \sigma_i^2}{s_n^2} \leq \epsilon^2 + \frac{1}{s_n^2} \sum_{i=1}^n \mathbb{E}\{Y_i^2 \mathbb{I}(|Y_i| \leq \epsilon s_n)\}.$$

Taking the limit as $n \rightarrow \infty$, under (A_1) :

$$\lim_{n \rightarrow \infty} \frac{\max_{1 \leq i \leq n} \sigma_i^2}{s_n^2} \leq \epsilon^2.$$

■

Proposition 4.3. Condition (A_2) implies *uniform asymptotic negligibility*:

$$\lim_{n \rightarrow \infty} \max_{1 \leq i \leq n} \mathbb{P}(|Y_i| > \epsilon s_n) = 0. \quad (A_3)$$

◆

Proof. By Chebyshev's inequality:

$$\lim_{n \rightarrow \infty} \max_{1 \leq i \leq n} \mathbb{P}(|Y_i| > \epsilon s_n) \leq \lim_{n \rightarrow \infty} \max_{1 \leq i \leq n} \frac{\mathbb{E}(Y_i^2)}{\epsilon^2 s_n^2} = \frac{1}{\epsilon^2} \lim_{n \rightarrow \infty} \frac{\max_{1 \leq i \leq n} \sigma_i^2}{s_n^2} = 0.$$

■

Proposition 4.4. The **Lyapunov condition** requires that $\exists \delta > 0$ such that:

$$\lim_{n \rightarrow \infty} \frac{\sum_{i=1}^n \mathbb{E}|Y_i|^{2+\delta}}{s_n^{2+\delta}} = 0. \quad (A_0)$$

The Lyapunov condition implies the *Lindeberg condition* (A_1) .

◆

Proof.

$$\begin{aligned} \frac{1}{s_n^2} \sum_{i=1}^n \mathbb{E}\{Y_i^2 \mathbb{I}(|Y_i| > \epsilon s_n)\} &= \sum_{i=1}^n \mathbb{E}\left\{\left|\frac{Y_i}{s_n}\right|^2 \cdot 1 \cdot \mathbb{I}(|Y_i|/(\epsilon s_n) > 1)\right\} \\ &\leq \sum_{i=1}^n \mathbb{E}\left\{\left|\frac{Y_i}{s_n}\right|^2 \left|\frac{Y_i}{s_n}\right|^\delta \mathbb{I}(|Y_i|/(\epsilon s_n) > 1)\right\} \\ &\leq \frac{1}{\epsilon^\delta} \frac{\sum_{i=1}^n \mathbb{E}|Y_i|^{2+\delta}}{s_n^{2+\delta}} \rightarrow 0. \end{aligned}$$

■

Remark 4.3. Based on the preceding, the Lyapunov implies Lindeberg implies uniform asymptotic negligibility:

$$A_0 \implies A_1 \implies A_2 \implies A_3.$$

The Lindeberg condition implies asymptotic normality:

$$A_0 \implies A_1 \implies B.$$

Feller's theorem states that, under condition (A_2) , asymptotic normality holds if and only if the Lindeberg condition holds:

$$A_2 \implies (A_1 \iff B).$$

◆

4.3 Triangular Array

Theorem 4.3 (Triangular Array CLT). Suppose $(X_{n,i})$ is a sequence of independent random variables, $Y_{n,i} = X_{n,i} - \mathbb{E}(X_{n,i})$, $\text{Var}(X_{n,i}) = \mathbb{E}(Y_{n,i}^2) = \sigma_{n,i}^2 < \infty$, and $s_n^2 = \sum_{i=1}^{r_n} \sigma_{n,i}^2$. If, for $\forall \epsilon > 0$, the Lindeberg condition holds:

$$\lim_{n \rightarrow \infty} \frac{1}{s_n^2} \sum_{i=1}^{r_n} \mathbb{E}\{Y_{n,i}^2 \mathbb{I}(|Y_{n,i}| > \epsilon s_n)\} = 0,$$

then:

$$\frac{\sum_{i=1}^{r_n} (X_{n,i} - \mu_{n,i})}{\sqrt{(\sum_{i=1}^{r_n} \sigma_{n,i}^2)}} \xrightarrow{d} N(0, 1).$$

□

Remark 4.4. See Billingsley (1995), theorem 27.2.

◆

4.4 Delta Method

Theorem 4.4. Suppose r_n is a sequence of positive constants such that $r_n \rightarrow \infty$ as $n \rightarrow \infty$, and that:

$$r_n(T_n - \theta) \xrightarrow{d} X.$$

If g is continuously differentiable in a neighborhood of θ , then:

$$r_n\{g(T_n) - g(\theta)\} \xrightarrow{d} \dot{g}(\theta)X.$$

□

Proof. Since $r_n(T_n - \theta)$ converges in distribution, $r_n(T_n - \theta) = \mathcal{O}_p(1)$ or:

$$T_n - \theta = \mathcal{O}_p(r_n^{-1}).$$

Since $r_n \rightarrow \infty$, this shows $T_n = \theta + o_p(1)$. Take the Taylor expansion of g about θ :

$$\begin{aligned} g(T_n) &= g(\theta) + \dot{g}(\theta)(T_n - \theta) + \mathcal{O}_p(r_n^{-2}), \\ r_n\{g(T_n) - g(\theta)\} &= \dot{g}(\theta)r_n(T_n - \theta) + \mathcal{O}_p(r_n^{-1}). \end{aligned}$$

By Slutsky's theorem:

$$r_n\{g(T_n) - g(\theta)\} \xrightarrow{d} \dot{g}(\theta)X.$$

■

Example 4.1. Suppose that:

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} N(0, \boldsymbol{\Sigma}).$$

Let $g : \mathbb{R}^p \rightarrow \mathbb{R}$ denote a continuous differentiable mapping, then:

$$\sqrt{n}\{g(\hat{\boldsymbol{\theta}}_n) - g(\boldsymbol{\theta}_0)\} \xrightarrow{d} N\{0, \dot{g}(\boldsymbol{\theta}_0)' \boldsymbol{\Sigma} \dot{g}(\boldsymbol{\theta}_0)\},$$

where:

$$\dot{g}(\boldsymbol{\theta}_0) = \left. \frac{\partial g(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0}.$$



Remark 4.5. See also van der Vaart (1998), theorem 3.1.

